

PASSWORD STRENGTH CHECKER

Project Report

Name: Shivam Raj

College: Government Polytechnic Sonipat

Email: shivamgaming920966453@gmail.com

Phone: 7070317887

Title

Password Strength Checker Using Python

Project Name

Password Strength Checker

Objective

To build a Python application that checks password strength.

Problem Statement

Weak passwords can lead to unauthorized access and cyber attacks.

Research Findings

Strong passwords should include uppercase, lowercase, numbers, special characters and be at least 8 characters long.

Methodology

Input password → Validate rules → Calculate score → Display Weak/Medium/Strong.

Tools Used

Python 3, VS Code/IDLE, re module.

Implementation

The application evaluates password strength using regular expressions.

Screenshots

Insert screenshots of code and outputs here.

Results

The program successfully classifies passwords as Weak, Medium, or Strong.

Challenges

Regex implementation and validation logic.

Future Scope

GUI, password suggestions, crack-time estimation.

Conclusion

The project improves awareness of password security.

Python Source Code

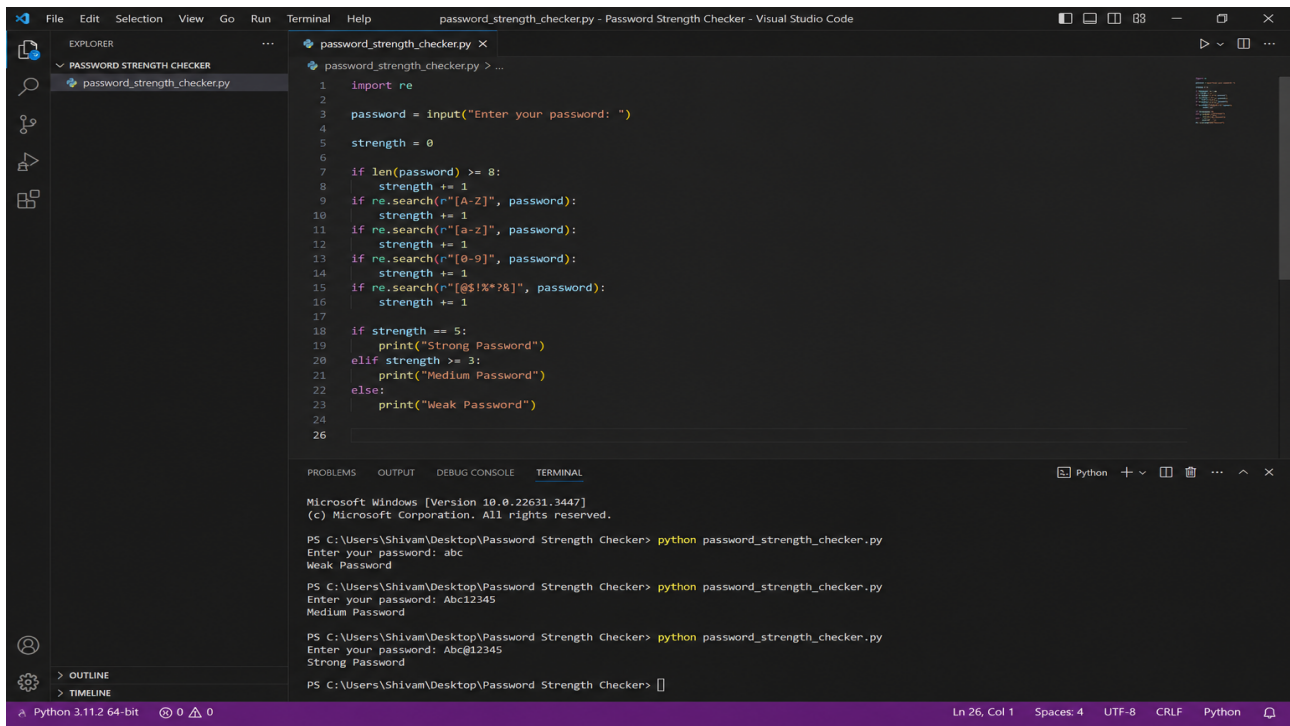
```
import re

password=input("Enter Password: ")
score=0
if len(password)>=8: score+=1
if re.search(r"[A-Z]",password): score+=1
if re.search(r"[a-z]",password): score+=1
if re.search(r"[0-9]",password): score+=1
if re.search(r"[@$!%*?&]",password): score+=1

if score==5:
    print("Strong Password")
elif score>=3:
    print("Medium Password")
else:
    print("Weak Password")
```

Password Strength Checker - Screenshot

Program execution screenshot:



The screenshot displays the Visual Studio Code interface with a Python script named `password_strength_checker.py` open in the editor. The script uses regular expressions to check password strength based on length, uppercase letters, lowercase letters, digits, and special characters. The terminal window shows the execution of the script for three different passwords: 'abc' (Weak Password), 'Abc12345' (Medium Password), and 'Abc@12345' (Strong Password).

```
1 import re
2
3 password = input("Enter your password: ")
4
5 strength = 0
6
7 if len(password) >= 8:
8     strength += 1
9 if re.search(r"[A-Z]", password):
10    strength += 1
11 if re.search(r"[a-z]", password):
12    strength += 1
13 if re.search(r"[0-9]", password):
14    strength += 1
15 if re.search(r"[@!%*?&]", password):
16    strength += 1
17
18 if strength == 5:
19     print("Strong Password")
20 elif strength >= 3:
21     print("Medium Password")
22 else:
23     print("Weak Password")
24
25
26
```

Terminal Output:

```
Microsoft Windows [Version 10.0.22631.3447]
(c) Microsoft Corporation. All rights reserved.

PS C:\Users\Shivam\Desktop>Password Strength Checker> python password_strength_checker.py
Enter your password: abc
Weak Password

PS C:\Users\Shivam\Desktop>Password Strength Checker> python password_strength_checker.py
Enter your password: Abc12345
Medium Password

PS C:\Users\Shivam\Desktop>Password Strength Checker> python password_strength_checker.py
Enter your password: Abc@12345
Strong Password

PS C:\Users\Shivam\Desktop>Password Strength Checker>
```